IoT Based Baby Monitoring System & Smart Cradle Baby Crying Detector

This project report is submitted for the fulfilment of the requirement for the course ECE 3216 of Bachelor of Science Degree in the Biomedical Engineering.

Submitted By

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Abstract

Nowadays, automation in the baby cradle system proved reliable and supports in effective baby monitoring. This system monitors the baby's mattresses condition, cry detection, body temperature etc., and thus provides better care for the babies. Also, it makes the parents very easy to take care of the baby and experience those worry-free feelings. Especially for the professional women who are busy in their work schedules, always find scarcity of time to take care of the baby. As a concern of money, they are unable to afford a maid or nursery to look after their baby. In this project, a local market is studied to decide the degree of automation and best suit mobile application required for the smart baby cradle prototype. Three sensors are used such as a sound sensor for detecting the crying sound of a baby, a rain sensor for detecting the condition of mattresses, and a temperature sensor for detecting the body temperature. To make this automated cradle internet of things (IoT) based, a Telegram platform is used and this entire feature is monitored and controlled via mobile phone. An automatic swing motion of the cradle is provided using a servo connecting with sound sensor. Automation in the cradle allows the remote parent to take care of their baby, monitor its actions, and maintain the balance in their professional and personal lives.

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1.0 Objectives:

The objectives of this project are as follows-

- To design a smart automated cradle by detecting baby's cry.
- To monitor baby's body temperature, diaper wetness.
- To apply Internet of Thing (IoT) for this monitor system by using Telegram application to notify parents.

2.0 Introduction:

In the present days, where both parents are busy in their professional life, it has become veritably tough for them to get enough time to look after their babies. It has been seen that most of the time babies stop crying or sleeping when they are in the cradle due to giving them a gentle swing movement. In the here and now, it's problematic for parents and nurses to sit close to their children and soothe them whenever they cry. In such a situation, parents fail to balance their duties at work and parenthood. So IoT-based automation of the baby cradle is carried out, which would support the parents and homemakers during their domestic work to look after their babies with paying the least attention. Also, it is a reality that professional women find it very tough to concentrate on her children due to busy house life schedule. The situation becomes more critical when her job or domestic business does not allow to compromise for attending her child's needs. That is why the automated cradle is the correct answer for it.

The automatic baby cradle system is not yet significantly common in developing countries where local manufacturers dominate the cradle demand. The targeted customer comes from families where members spend plenty of time with the baby.

The automatic baby cradle available in the market currently uses the DC motor to swing the baby buggy. The motor consumes more power and produces noise, which disturbs the baby and reduces the comfort degree of the baby. Whereas occasionally starting off, the baby cradle has to jerk full swing.

The introductory factors used in the current baby cradle are controller, motor, power supply, timer, and different sensors like sound detector, wet detector, temperature detector, etc. The timer is set to the controller which provides the movement to the cradle for a previous set time. The presented automation in the cradle is intended to furnish the following features as to reduce the labor of parents as well as take care of the baby, to control the system via the mobile app, to determine temperature, mattresses humidity, and crying voice of the baby, to operate

the cradle as per the program specified, etc. This proposed cradle is to serve not only for domestic application but at mass-utility places like daycare centers, hospitals, etc. most conveniently and affordably.

3.0 Literature review:

To overcome the challenges associated with smart cradle by detecting baby crying, many works have been done to develop this monitoring system.

3.1 IOT Based Baby Monitoring System

IoT-based baby monitoring has been one of the main points of discussion in recent years. One of the earliest works in this field proposed an integrated system of sensors to monitor critical baby health parameters such as temperature, heart rate, and motion, whose data was to be transmitted on a wireless network [1]. Application of these systems to real-time healthcare monitoring has shown that they are capable of continuous data provision to caregivers and thus enhancing infant safety [2]. Recent innovations in this technology include the integration of facial recognition and cry detection algorithms that can give more accurate assessments about the health needs of the baby. Further, real-time feedback may drastically reduce response time towards the health concerns of the baby [3][4].

3.2 Cry Detection Using Machine Learning

Cry detection in modern baby monitors is one of the most in-demand features invested in by high-end manufacturers of such system .Machine learning models are trained using large datasets of recorded baby cries, which are analyzed to identify patterns and correlate them with potential causes [5]. Of importance, the approaches developed include the classification and prediction of a cry using neural networks to assist the system in informing caregivers of their findings with more specificity regarding the need of the baby [6][7]. It increased a lot in terms of the way the system could respond to a real-time infant's needs.

3.3 Sensor Integration in Baby Monitoring System

Sensor integration has been one of the active research areas for IoT-based baby monitoring systems. The most common ones include temperature sensors, motion sensors, and heart rate sensors to track the health status of an infant. Wearable sensors, which monitor

constantly, have recently been introduced to always offer a continuous assessment of the physical status of the infant [8]. The accelerometers and gyroscopes will also be allowed to monitor the position of the baby, crucial for ensuring that the baby's safety while in sleep or in motion is catered for, according to [9].It will be easy for these sensors, once integrated with IoT platforms, to allow the caregiver to have real-time warnings where any reading will fall out of the safety threshold, ensuring prompt reaction in potential health threats [10][11].

3.4 Socio-Economic Impact of Baby Monitoring Systems

The socio-economic impact of IoT-based baby monitoring systems exceeds the technology itself to dive deep into the area of experience of caregiving. Most of the caregivers, especially working ones or those managing a number of children, live under remarkable stress and anxiety due to constant monitoring of the well-being of their infant. IoT systems minimize such effort by facilitating remote monitoring and creating peace of mind, thus enhancing the overall caregiving experience [12][13].

Moreover, such systems may enable the caregiver to save some money by avoiding unnecessary visits to the hospital. In this respect, early detection of health problems may enable a caregiver to stay away from emergency room visits, thus bringing healthcare costs down[14]. IoT-based baby-monitoring systems are thus an economically viable solution for every family, rather than just a technological advancement.

3.5 Current Trends in the Field of Baby Monitoring System

The emergent usages of IoT technologies reflect modern trends in integration with artificial intelligence and machine learning within the sphere of baby monitoring. Such a baby monitoring system does not confine its capabilities to only simple monitoring but rather predictively creates insights about future risks of health conditions from previous historical data. The AI-powered systems can analyze the data collected from multiple sensors to predict health issues, such as fever or dehydration, before they become serious problems, as identified in [15] and [16]. This kind of predictive capability will revolutionize infant care by affording much more proactive approaches toward health management.

4.0 Project description:

4.1 Arduino Uno Microcontroller:

The Arduino UNO is the best board to get started with electronics and coding. If this is your first experience tinkering with the platform, the UNO is the most robust board you can start playing with. The UNO is the most used and documented board of the whole Arduino family. Arduino UNO is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started[17].

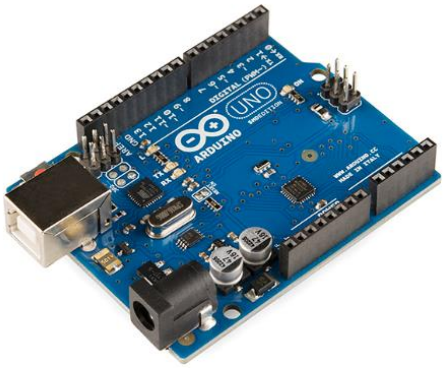


Figure 4.1.1: Arduino Uno

4.2 Node MCU ESP8266

ESP8266 is an open-source firmware, hardware, and software development board with ESP8266 Wi-fi SoC (System on Chip) chip based on LUA. It is a cost-effective, compact, and useful board for IoT applications. It is compatible with Arduino IDE and Python[18].



Figure 4.2.1: Node MCU ESP8266

4.3 Sound Sensor

Sound sensors are electronic devices that are used to detect and measure sound waves. They are widely used in a variety of applications, including sound recording, noise measurement, and speech recognition. In this blog post, we will discuss the working principle, features, and applications of sound sensors[19].

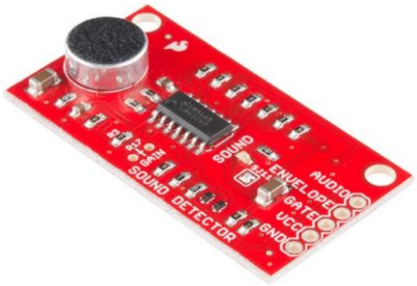


Figure 4.3.1: Sound Sensor

4.4 Servo Motor

A servo motor is a self-contained electrical device that moves parts of a machine with high efficiency and great precision. In simpler terms, a servo motor is a BLDC motor with a sensor for positional feedback. This allows the output shaft to be moved to a particular angle, position, and velocity that a regular motor cannot do. However, a servo motor is only one part of a closed-loop motion control system. A complete motion system includes an amplifier, control circuit, drive gears, potentiometer, shaft, and either an encoder or resolver as well as the servo motor [20].



Figure 4.4.1:MG996R 9g Servo Motor

4.5 Veroboard

Veroboard is a brand of stripboard, a pre-formed circuit board material of copper strips on an insulating bonded paperboard which was originated. It was introduced as a general-purpose material for use in constructing electronic circuits – differing from purpose-designed printed circuit boards (PCBs) in that a variety of electronic circuits may be constructed using a standard wiring board. Thickness of a Single Sided Veroboard is about 1.3mm and hole diameter is 1.0mm.



Figure 4.5.1: Veroboard

4.6 Li-ion Rechargeable Battery

A lithium-ion battery is a type of rechargeable battery that uses lithium ions as the primary component of its electrolyte. During the discharge cycle, lithium atoms in the anode are ionized and separated from their electrons. The lithium ions move from the 9 anode and pass through the electrolyte until they reach the cathode, where they recombine with their electrons and electrically neutralize [21].



Figure 4.6.1: 3.7 V Li-ion Rechargeable Battery

4.7 NTC Thermistor

Thermistors are temperature-sensing elements made of semiconductor material that has been sintered to display large changes in resistance in proportion to small changes in

temperature. These solid-state temperature sensors act like electrical resistors that are temperature sensitive. That is where the name, a clear combination of the words thermal and resistor, comes from. Ametherm specializes in NTC, or negative temperature coefficient, thermistors.



Figure 4.7.1: NTC Thermistor



Figure 4.7.2: External Temperature Probe

4.8 Rain Sensor

A rain sensor or rain switch is a switching device activated by rainfall. Here we use it to detect urine.



Figure 4.8.1: Rain Sensor

4.9 Circuit Diagram

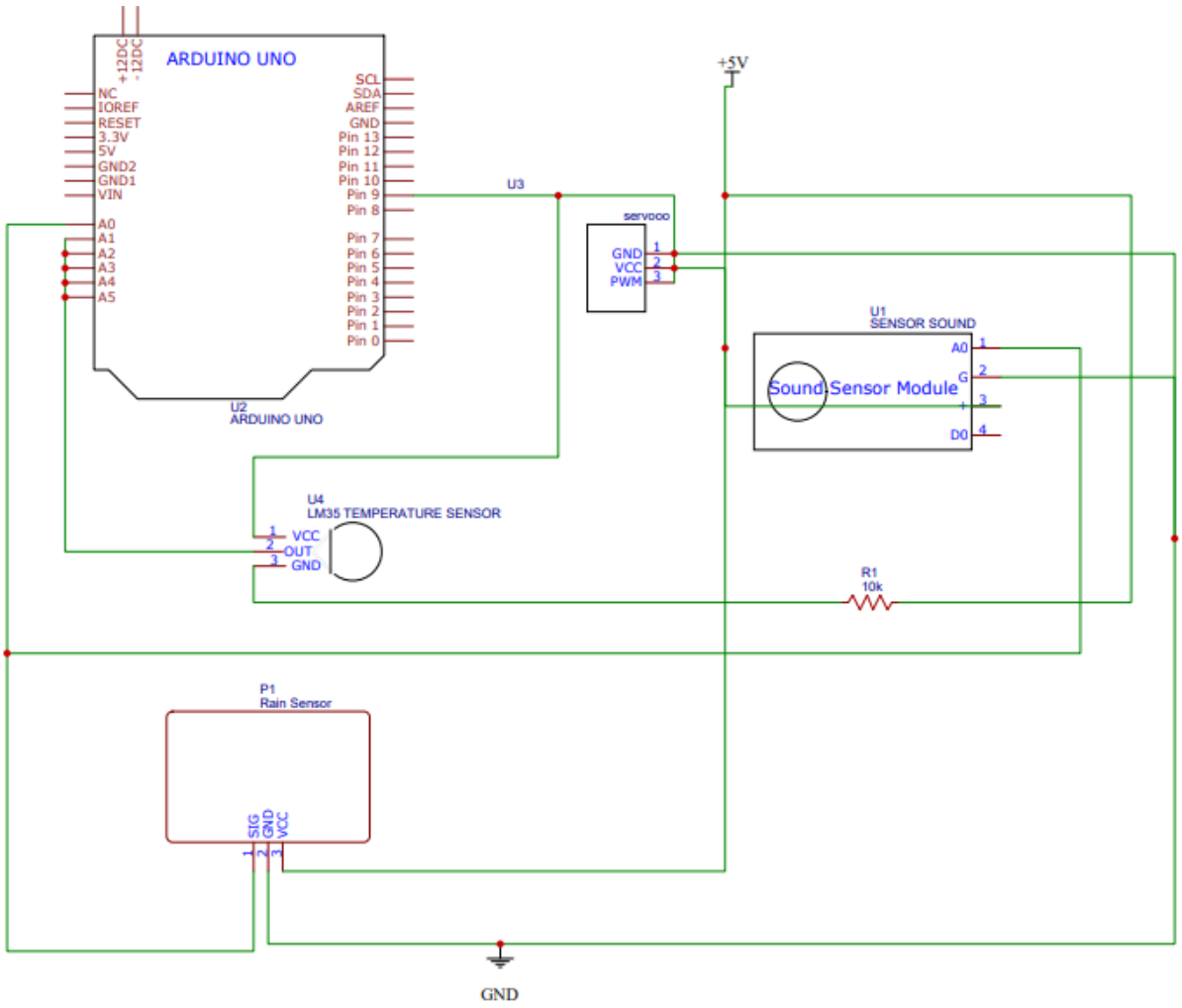


Figure 4.9.1: Circuit Diagram using EasyEDA tool

4.10 Assembled hardware

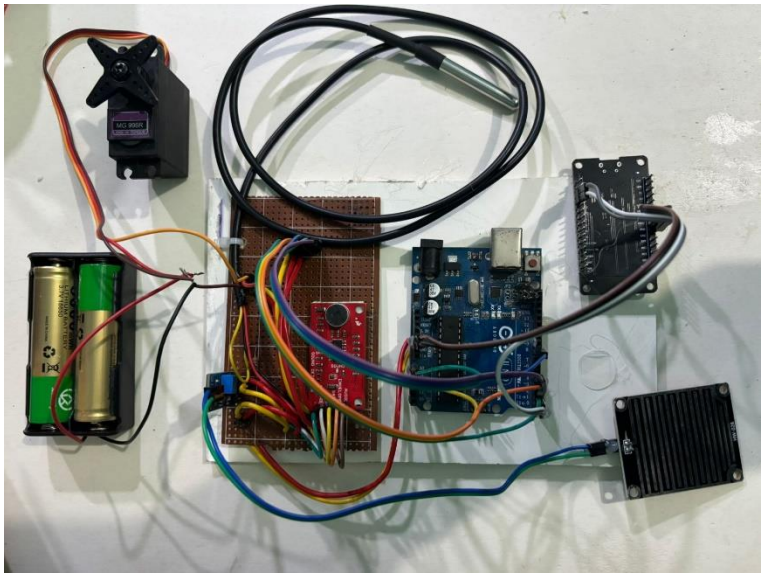


Figure 4.10.1: Assembled Circuit



Figure 4.10.2 : Makeshift cradle setup for demonstration

5.0 Advantages over pre-existing methods

5.1 Cost Effective Solution

The proposed smart cradle costs approximately 3721 BDT, significantly cheaper than the Electronic baby cradle ($22.08-25.68).

5.2 Enhanced Accessibility

Combines advanced functionality with real-time monitoring at a reduced price point.

5.3 Practical Design

Focuses on affordability without compromising quality or usability, making it accessible for easier applications.

6.0 Result and analysis

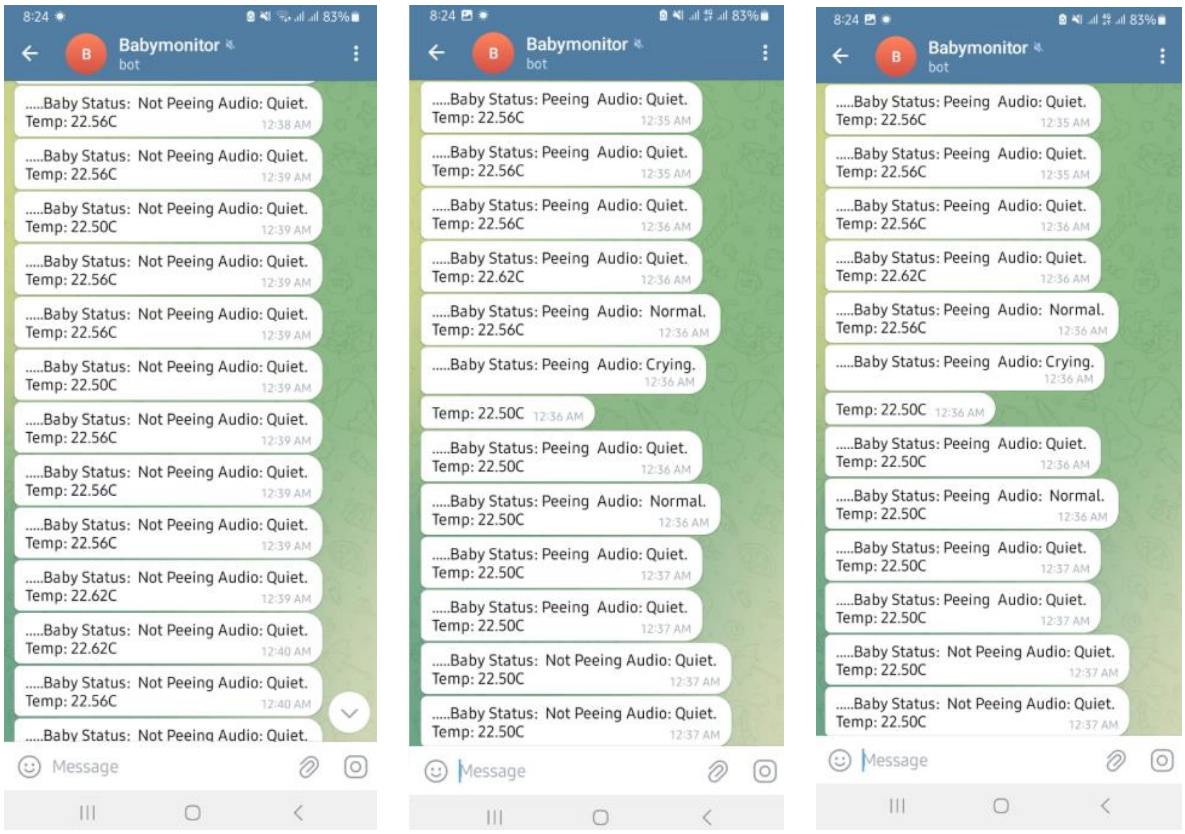


Figure 6.0.1: Received data on Telegram application

The smart baby cradle system was tested to evaluate its performance in real-time monitoring of a baby's condition. The system integrates a sound sensor, rain sensor, and temperature sensor with a Telegram bot interface to send status updates directly to parents. The results indicate that the system successfully detects various activities and transmits the data in real-time via the Telegram bot.

Table 6.0.1: Status Messages and System Functionality of the Smart Baby Cradle

Parameter	Status Message	Description
Baby Cry Detection	Audio: Crying/ Audio: Quite	Detects when the baby cries and starts the cradle rocking automatically.
Urination Detection	Peeing/Not Peeing	Detects if the baby has urinated using a rain sensor and sends notification through Telegram bot..
Temperature Reading	Temp: 22.50℃	Continuously monitors the surrounding temperature and displays it on the Telegram bot.

The system ensures automated actions and remote monitoring, reducing the need for continuous physical supervision. The results demonstrate that the smart cradle is a reliable, efficient solution for baby monitoring, offering convenience and peace of mind to parents.

7.0 Cost analysis:

Table 7.0.1: Components cost of IoT Based Baby Monitoring System & Smart Cradle Baby Crying Detector:

Component	Quantity	Price (BDT)
Arduino Uno	1	995
Rain Drop Sensor	1	80
SparkFun Sound Detector	1	1582
NodeMCU ESP8266	1	420
NTC Thermistor Temperature Sensor	1	100
Veroboard	1	35
Servo Motor	1	209
Li-ion Rechargeable Battery (3.7 V)	2	100
Jumper wires	-	100
Miscellaneous	-	100
Total		3721

8.0 Conclusion

The proposed design has successfully detected simulated cry of a baby to rock the makeshift cradle rock gently with the servo rotation mechanism . Additionally, the system detected the temperature and the wetness. The information could successfully be sent to Telegram application ensuring that the parents can keep an eye on their baby even when they are not present in the same room with the baby.

Future scopes:

- ✓ The design has the potential to be a cost-effective version of the currently costly smart cradles.
- ✓ The design could be modified to monitor sick babies by adding certain sensors that could keep track of their vital signs.
- ✓ This system can also be incorporated with a baby monitoring camera and the IoT approach opens many gates to enhance the data sending capability prioritizing the safety and comfort of the babies.

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APPENDIX

Code:

```
#include <Wire.h>
#include <Adafruit_BMP085.h>
#define seaLevelPressure_hPa 1013.25
#include<LiquidCrystal_I2C.h>
#include<SoftwareSerial.h>


SoftwareSerial BT(10,11);
LiquidCrystal_I2C lcd(0x27,16,2);
Adafruit_BMP085 bmp;


void setup() {
  Serial.begin(9600);
  BT.begin(9600);
  if (!bmp.begin()) {
        Serial.println("Could not find a valid BMP085 sensor, check wiring!");
        while (1) {}
  }
  pinMode(A0,INPUT);
  lcd.init();
  lcd.backlight();
  lcd.clear();
}

void loop() {
    Serial.print("Temperature = ");
    Serial.print(bmp.readTemperature());
    Serial.println(" *C");
lcd.setCursor(0,0);
lcd.print("T:");
lcd.print(bmp.readTemperature());
lcd.print("C");
    Serial.print("Pressure = ");
    Serial.print(bmp.readPressure());
    Serial.println(" Pa");

    int x=map(bmp.readPressure(),101350,104090,80,110);
    Serial.println(x);
lcd.print(" P:");
lcd.print(x);
lcd.print("Hg");

    int y=analogRead(A0);
    int ym=map(y,200,800,60,100);
    Serial.println(ym);
```

```
   lcd.setCursor(0,1);
lcd.print("H:");
lcd.print(ym);
lcd.print("bpm");


BT.print(bmp.readTemperature());
BT.print(",");
BT.print(x);
BT.print(",");
BT.print(ym);
BT.print(";");


   Serial.println();

   delay(1000);
}
```